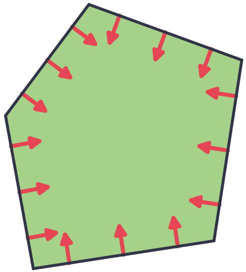
Boundary forces in generalized density functional theories

Chih-Chun Wang, Markus Penz, Christian Schilling
April 21, 2026

Workshop on the Foundations and Extensions of Density Functional Theory



Motivation & Outline

- Diverging repulsive force from boundary of functional domain in fermionic & bosonic RDMFT

Schilling, C., Schilling, R. Diverging Exchange Force and Form of the Exact Density Matrix Functional. Phys. Rev. Lett. (2019)
Benavides-Riveros, C. L., et al. Reduced Density Matrix Functional Theory for Bosons. Phys. Rev. Lett. (2020)

Motivation & Outline

- Diverging repulsive force from boundary of functional domain in fermionic & bosonic RDMFT
  - Dynamic correlations in quantum chemistry

Schilling, C., Schilling, R. Diverging Exchange Force and Form of the Exact Density Matrix Functional. Phys. Rev. Lett. (2019)
Benavides-Riveros, C. L., et al. Reduced Density Matrix Functional Theory for Bosons. Phys. Rev. Lett. (2020)

Motivation & Outline

- Diverging repulsive force from boundary of functional domain in fermionic & bosonic RDMFT
  - Dynamic correlations in quantum chemistry
  - Exact properties of the functional

Schilling, C., Schilling, R. Diverging Exchange Force and Form of the Exact Density Matrix Functional. Phys. Rev. Lett. (2019)
Benavides-Riveros, C. L., et al. Reduced Density Matrix Functional Theory for Bosons. Phys. Rev. Lett. (2020)

Motivation & Outline

- Diverging repulsive force from boundary of functional domain in fermionic & bosonic RDMFT
  - Dynamic correlations in quantum chemistry
  - Exact properties of the functional

Goal: understand and formalize boundary behavior in general setting

Schilling, C., Schilling, R. Diverging Exchange Force and Form of the Exact Density Matrix Functional. Phys. Rev. Lett. (2019)
Benavides-Riveros, C. L., et al. Reduced Density Matrix Functional Theory for Bosons. Phys. Rev. Lett. (2020)

Motivation & Outline

- Diverging repulsive force from boundary of functional domain in fermionic & bosonic RDMFT
  - Dynamic correlations in quantum chemistry
  - Exact properties of the functional

Goal: understand and formalize boundary behavior in general setting

Outline:

Schilling, C., Schilling, R. Diverging Exchange Force and Form of the Exact Density Matrix Functional. Phys. Rev. Lett. (2019)
Benavides-Riveros, C. L., et al. Reduced Density Matrix Functional Theory for Bosons. Phys. Rev. Lett. (2020)

Motivation & Outline

- Diverging repulsive force from boundary of functional domain in fermionic & bosonic RDMFT
  - Dynamic correlations in quantum chemistry
  - Exact properties of the functional

Goal: understand and formalize boundary behavior in general setting

Outline:

1. Generalized density functional theory framework

Schilling, C., Schilling, R. Diverging Exchange Force and Form of the Exact Density Matrix Functional. Phys. Rev. Lett. (2019)
Benavides-Riveros, C. L., et al. Reduced Density Matrix Functional Theory for Bosons. Phys. Rev. Lett. (2020)

Motivation & Outline

- Diverging repulsive force from boundary of functional domain in fermionic & bosonic RDMFT
  - Dynamic correlations in quantum chemistry
  - Exact properties of the functional

Goal: understand and formalize boundary behavior in general setting

Outline:

1. Generalized density functional theory framework
2. Boundary force theorem & proof sketch

Schilling, C., Schilling, R. Diverging Exchange Force and Form of the Exact Density Matrix Functional. Phys. Rev. Lett. (2019)
Benavides-Riveros, C. L., et al. Reduced Density Matrix Functional Theory for Bosons. Phys. Rev. Lett. (2020)

Motivation & Outline

- Diverging repulsive force from boundary of functional domain in fermionic & bosonic RDMFT
  - Dynamic correlations in quantum chemistry
  - Exact properties of the functional

Goal: understand and formalize boundary behavior in general setting

Outline:

1. Generalized density functional theory framework
2. Boundary force theorem & proof sketch
3. Application: recovering the interaction from boundary information

Schilling, C., Schilling, R. Diverging Exchange Force and Form of the Exact Density Matrix Functional. Phys. Rev. Lett. (2019)
Benavides-Riveros, C. L., et al. Reduced Density Matrix Functional Theory for Bosons. Phys. Rev. Lett. (2020)

Generalized density functional theory

Definition

A *functional theory* is a tuple $(\mathcal{H}, V, \iota, H_0)$, where

Generalized density functional theory

Definition

A *functional theory* is a tuple $(\mathcal{H}, V, \iota, H_0)$, where

- $\mathcal{H}$ is a finite-dimensional Hilbert space

Generalized density functional theory

Definition

A *functional theory* is a tuple $(\mathcal{H}, V, \iota, H_0)$, where

- $\mathcal{H}$ is a finite-dimensional Hilbert space
- V is a finite-dimensional vector space over $\mathbb{R}$

Generalized density functional theory

Definition

A *functional theory* is a tuple $(\mathcal{H}, V, \iota, H_0)$, where

- $\mathcal{H}$ is a finite-dimensional Hilbert space
- V is a finite-dimensional vector space over $\mathbb{R}$
- $\iota : V \to \mathrm{Herm}(\mathcal{H})$ is a linear map

Generalized density functional theory

Definition

A *functional theory* is a tuple $(\mathcal{H}, V, \iota, H_0)$, where

- $\mathcal{H}$ is a finite-dimensional Hilbert space
- V is a finite-dimensional vector space over $\mathbb{R}$
- $\iota : V \to \mathrm{Herm}(\mathcal{H})$ is a linear map
- $H_0 \in \mathrm{Herm}(\mathcal{H})$ is an arbitrary but fixed “interaction” operator.

Generalized density functional theory

Definition

A *functional theory* is a tuple $(\mathcal{H}, V, \iota, H_0)$, where

- $\mathcal{H}$ is a finite-dimensional Hilbert space
- V is a finite-dimensional vector space over $\mathbb{R}$
- $\iota : V \to \mathrm{Herm}(\mathcal{H})$ is a linear map
- $H_0 \in \mathrm{Herm}(\mathcal{H})$ is an arbitrary but fixed "interaction" operator.

Idea: want to study family of Hamiltonians $H(v) := \iota(v) + H_0$.

Generalized density functional theory

Definition

A *functional theory* is a tuple $(\mathcal{H}, V, \iota, H_0)$, where

- $\mathcal{H}$ is a finite-dimensional Hilbert space
- V is a finite-dimensional vector space over $\mathbb{R}$
- $\iota : V \to \mathrm{Herm}(\mathcal{H})$ is a linear map
- $H_0 \in \mathrm{Herm}(\mathcal{H})$ is an arbitrary but fixed “interaction” operator.

Idea: want to study family of Hamiltonians $H(v) := \iota(v) + H_0$.

Definition

The *density map* is $\mu := \iota^* : \mathbb{P}(\mathcal{H}) \to V^*$. Equivalently:

$$\langle \mu[\psi], v \rangle = \langle \psi, \iota(v)\psi \rangle$$

Examples

DFT

- $V = \mathbb{R}^d$
- $\mathcal{H} = \bigwedge^N(\mathcal{H}_1)$ or $\mathrm{Sym}^N(\mathcal{H}_1)$
- $\iota(v) = \sum_{i=1}^d v_i a_i^* a_i$
- $\mu : [\psi] \mapsto \rho$

Examples

DFT

- $V = \mathbb{R}^d$
- $\mathcal{H} = \bigwedge^N(\mathcal{H}_1)$ or $\mathrm{Sym}^N(\mathcal{H}_1)$
- $\iota(v) = \sum_{i=1}^d v_i a_i^* a_i$
- $\mu : [\psi] \mapsto \rho$

RDMFT

- $V = \mathrm{Herm}(\mathcal{H}_1)$
- $\mathcal{H} = \bigwedge^N(\mathcal{H}_1)$ or $\mathrm{Sym}^N(\mathcal{H}_1)$
- $\iota(v) = \sum_{i=1}^d v_{ij} a_i^* a_j$
- $\mu = N\,\mathrm{Tr}_{N-1}(\cdot)$ (1RDM)

Examples

DFT

- $V = \mathbb{R}^d$
- $\mathcal{H} = \bigwedge^N(\mathcal{H}_1)$ or $\mathrm{Sym}^N(\mathcal{H}_1)$
- $\iota(v) = \sum_{i=1}^d v_i a_i^* a_i$
- $\mu : [\psi] \mapsto \rho$

RDMFT

- $V = \mathrm{Herm}(\mathcal{H}_1)$
- $\mathcal{H} = \bigwedge^N(\mathcal{H}_1)$ or $\mathrm{Sym}^N(\mathcal{H}_1)$
- $\iota(v) = \sum_{i=1}^d v_{ij} a_i^* a_j$
- $\mu = N\,\mathrm{Tr}_{N-1}(\cdot)$ (1RDM)

Spins (abelian)

- $V = \mathbb{R}^N$
- $\mathcal{H} = (\mathbb{C}^2)^{\otimes N}$
- $\iota(v) = \sum_{i=1}^N v_i Z_i$
- $\mu : [\psi] \mapsto (\langle Z_1 \rangle, \cdots, \langle Z_N \rangle)$

Spins (nonabelian)

- $V = \mathbb{R}^{3N}$
- $\mathcal{H} = (\mathbb{C}^2)^{\otimes N}$
- $\iota(v) = \sum_{i=1}^N (v_i^x X_i + v_i^y Y_i + v_i^z Z_i)$
- $\mu : [\psi] \mapsto (\langle X_1 \rangle, \langle Y_1 \rangle, \langle Z_1 \rangle, \langle X_2 \rangle, \cdots)$

Constrained search functional

Definition

The *(pure state) constrained search functional* is

$$\begin{aligned} \mathcal{F} : \mu(\mathbb{P}(\mathcal{H})) &\to \mathbb{R} \\ \rho &\mapsto \min_{[\psi]\in\mu^{-1}(\rho)} \langle \psi, H_0 \psi \rangle \,. \end{aligned}$$

Constrained search functional

Definition

The *(pure state) constrained search functional* is

$$\begin{aligned} \mathcal{F} : \mu(\mathbb{P}(\mathcal{H})) &\to \mathbb{R} \\ \rho &\mapsto \min_{[\psi]\in\mu^{-1}(\rho)} \langle \psi, H_0\psi \rangle . \end{aligned}$$

The domain $\mu(\mathbb{P}(\mathcal{H}))$ is the *pure state joint numerical range* or *set of pure state (N-)representable densities*.

Abelian functional theories

We will focus on *abelian* functional theories:

Abelian functional theories

We will focus on *abelian* functional theories:

Definition

A functional theory is *abelian* if $\iota(V)$ is commutative.

Abelian functional theories

We will focus on *abelian* functional theories:

Definition

A functional theory is *abelian* if $\iota(V)$ is commutative.

Definition

A density $\omega \in V^*$ is a *weight* if

$$\exists \psi \neq 0 : \forall v \in V : \iota(v)\psi = \langle \omega, v \rangle \psi.$$

Abelian functional theories

We will focus on *abelian* functional theories:

Definition

A functional theory is *abelian* if $\iota(V)$ is commutative.

Definition

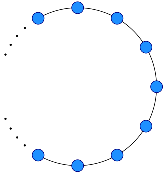
A density $\omega \in V^*$ is a *weight* if

$$\exists \psi \neq 0 : \forall v \in V : \iota(v)\psi = \langle \omega, v \rangle \psi.$$

Fact:

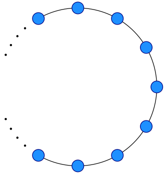
$$\mu(\mathbb{P}(\mathcal{H})) = \mathrm{conv}(\Omega) = \text{convex hull of all weights}$$

Example - bosons in momentum space



Wang, C.-C., Schilling C. A Geometric Approach to Strongly Correlated Bosons: From N-Representability to the Generalized BEC Force. Phys. Rev. B. (forthcoming)

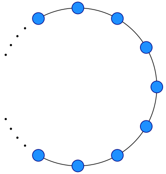
Example - bosons in momentum space



- $\mathcal{H} =$ momentum-P subspace of $\mathrm{Sym}^N(\mathbb{C}^d)$

Wang, C.-C., Schilling C. A Geometric Approach to Strongly Correlated Bosons: From N-Representability to the Generalized BEC Force. Phys. Rev. B. (forthcoming)

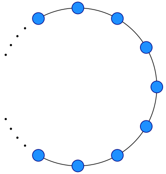
Example - bosons in momentum space



- $\mathcal{H} =$ momentum-P subspace of $\mathrm{Sym}^N(\mathbb{C}^d)$
- $V = \mathbb{R}^d$

Wang, C.-C., Schilling C. A Geometric Approach to Strongly Correlated Bosons: From N-Representability to the Generalized BEC Force. Phys. Rev. B. (forthcoming)

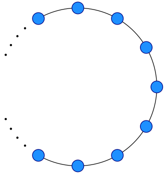
Example - bosons in momentum space



- $\mathcal{H} =$ momentum-P subspace of $\mathrm{Sym}^N(\mathbb{C}^d)$
- $V = \mathbb{R}^d$
- $\iota(v) = \sum_{k=0}^{d-1} v_k a_k^* a_k$

Wang, C.-C., Schilling C. A Geometric Approach to Strongly Correlated Bosons: From N-Representability to the Generalized BEC Force. Phys. Rev. B. (forthcoming)

Example - bosons in momentum space



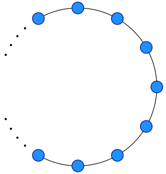
- $\mathcal{H}$ = momentum-P subspace of $\mathrm{Sym}^N(\mathbb{C}^d)$
- $V = \mathbb{R}^d$
- $\iota(v) = \sum_{k=0}^{d-1} v_k a_k^* a_k$

set of weights Ω

$$= \left\{ n \in \mathbb{N}^d \mid \sum_{k=0}^{d-1} n_k = N \quad \sum_{k=0}^{d-1} k n_k \equiv P \pmod d \right\}$$

Wang, C.-C., Schilling C. A Geometric Approach to Strongly Correlated Bosons: From N-Representability to the Generalized BEC Force. Phys. Rev. B. (forthcoming)

Example - bosons in momentum space

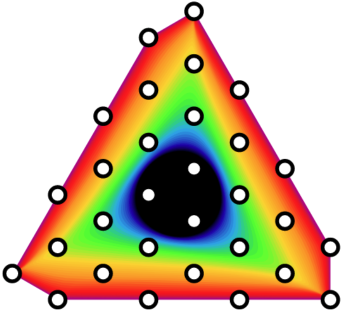


- $\mathcal{H} =$ momentum-P subspace of $\mathrm{Sym}^N(\mathbb{C}^d)$
- $V = \mathbb{R}^d$
- $\iota(v) = \sum_{k=0}^{d-1} v_k a_k^* a_k$

set of weights Ω

$$= \left\{ n \in \mathbb{N}^d \mid \sum_{k=0}^{d-1} n_k = N \quad \sum_{k=0}^{d-1} k n_k \equiv P \pmod{d} \right\}$$

$|11, 1, 0\rangle \nearrow$

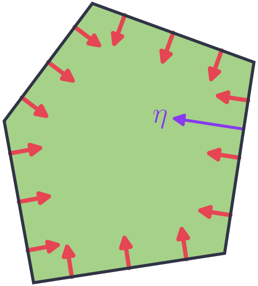


$d = 3, N = 12, P = 1$

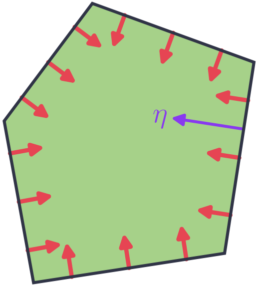
$H_0 =$ Hubbard on-site interaction

Wang, C.-C., Schilling C. A Geometric Approach to Strongly Correlated Bosons: From N-Representability to the Generalized BEC Force. Phys. Rev. B. (forthcoming)

Boundary forces



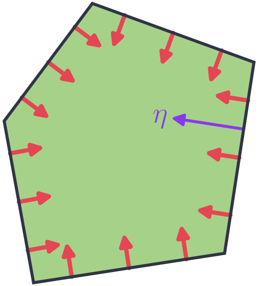
Boundary forces



$$\mathcal{F}(\rho_* + \epsilon\eta) \approx \mathcal{F}_0 - \sqrt{\epsilon}\mathcal{G}$$

Schilling, C., Schilling, R. Diverging Exchange Force and Form of the Exact Density Matrix Functional. Phys. Rev. Lett. (2019)
Maciążek, T. Repulsively Diverging Gradient of the Density Functional in the Reduced Density Matrix Functional Theory. New J. Phys. (2021)

Boundary forces



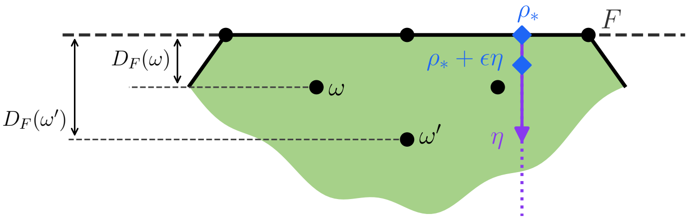
$$\mathcal{F}(\rho_* + \epsilon\eta) \approx \mathcal{F}_0 - \sqrt{\epsilon}\mathcal{G}$$

$$\frac{\mathrm{d}}{\mathrm{d}\sqrt{\epsilon}}\mathcal{F}(\rho_* + \epsilon\eta) = -\mathcal{G}$$

Schilling, C., Schilling, R. Diverging Exchange Force and Form of the Exact Density Matrix Functional. Phys. Rev. Lett. (2019)
Maciążek, T. Repulsively Diverging Gradient of the Density Functional in the Reduced Density Matrix Functional Theory. New J. Phys. (2021)

The boundary force formula

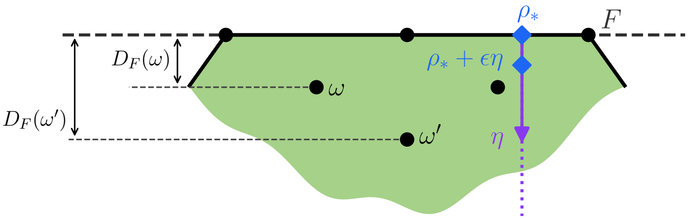
Pick:



The boundary force formula

Pick:

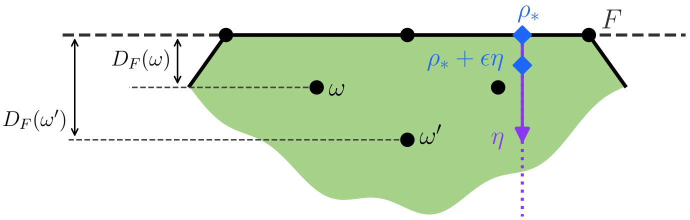
- facet $F \subset \mathrm{conv}(\Omega)$



The boundary force formula

Pick:

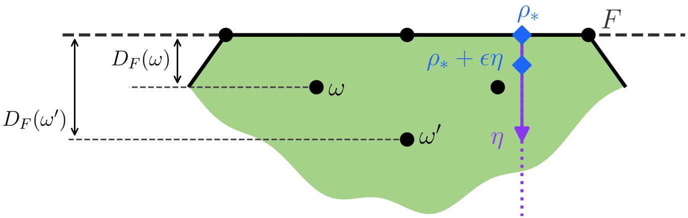
- facet $F \subset \mathrm{conv}(\Omega)$
- density $\rho_* \in F$



The boundary force formula

Pick:

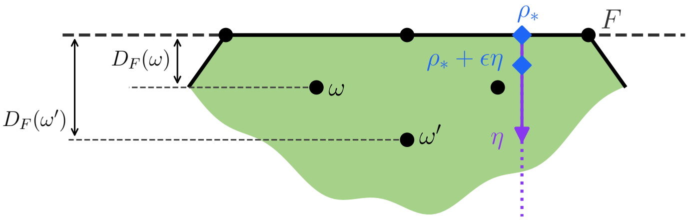
- facet $F \subset \mathrm{conv}(\Omega)$
- density $\rho_* \in F$
- inward vector $\eta \in V^*$



The boundary force formula

Pick:

- facet $F \subset \mathrm{conv}(\Omega)$
- density $\rho_* \in F$
- inward vector $\eta \in V^*$



Theorem

Let $\rho_* \in F$ be "regular". Then

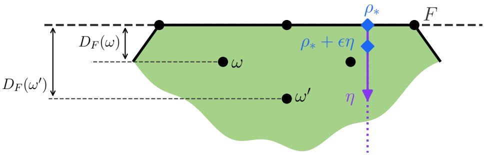
$$\lim_{\epsilon \downarrow 0} \frac{\mathcal{F}_p(\rho_* + \epsilon\eta) - \mathcal{F}_p(\rho_*)}{\sqrt{\epsilon}} = -2\sqrt{\sum_{\omega \notin \Omega_F} \frac{\|\Pi_\omega H_0 \Phi\|^2}{D_F(\omega)}}.$$

The boundary force formula

Theorem

Let $\rho_* \in F$ be "regular". Then

$$\lim_{\epsilon\downarrow 0} \frac{\mathcal{F}_p(\rho_* + \epsilon\eta) - \mathcal{F}_p(\rho_*)}{\sqrt{\epsilon}} = -2\sqrt{\sum_{\omega\notin\Omega_F} \frac{\|\Pi_\omega H_0 \Phi\|^2}{D_F(\omega)}}.$$

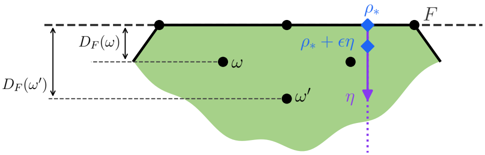


The boundary force formula

Theorem

Let $\rho_* \in F$ be "regular". Then

$$\lim_{\epsilon \downarrow 0} \frac{\mathcal{F}_p(\rho_* + \epsilon\eta) - \mathcal{F}_p(\rho_*)}{\sqrt{\epsilon}} = -2\sqrt{\sum_{\omega \notin \Omega_F} \frac{\|\Pi_\omega H_0 \Phi\|^2}{D_F(\omega)}}.$$



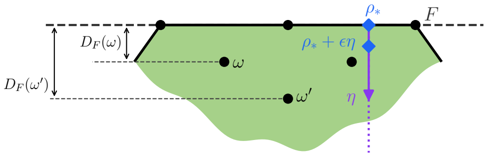
- $\Phi \in \mu^{-1}(\rho_*)$ minimizer of constrained search

The boundary force formula

Theorem

Let $\rho_* \in F$ be "regular". Then

$$\lim_{\epsilon \downarrow 0} \frac{\mathcal{F}_p(\rho_* + \epsilon\eta) - \mathcal{F}_p(\rho_*)}{\sqrt{\epsilon}} = -2\sqrt{\sum_{\omega \notin \Omega_F} \frac{\|\Pi_\omega H_0 \Phi\|^2}{D_F(\omega)}}.$$



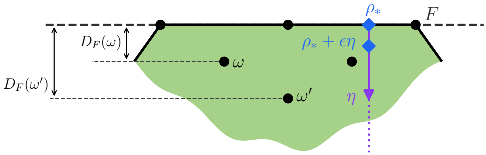
- $\Phi \in \mu^{-1}(\rho_*)$ minimizer of constrained search
- $\Pi_\omega : \mathcal{H} \to \mathcal{H}_\omega$ projection onto weight space

The boundary force formula

Theorem

Let $\rho_* \in F$ be "regular". Then

$$\lim_{\epsilon \downarrow 0} \frac{\mathcal{F}_p(\rho_* + \epsilon\eta) - \mathcal{F}_p(\rho_*)}{\sqrt{\epsilon}} = -2\sqrt{\sum_{\omega \notin \Omega_F} \frac{\|\Pi_\omega H_0 \Phi\|^2}{D_F(\omega)}}.$$



- $\Phi \in \mu^{-1}(\rho_*)$ minimizer of constrained search
- $\Pi_\omega : \mathcal{H} \to \mathcal{H}_\omega$ projection onto weight space
- $D_F(\omega)$ distance from ω to F

Proof idea

Proof idea

$$\mathcal{F}(\rho_*+\epsilon\eta) = \min_{[\psi]\in\mu^{-1}(\rho_*+\epsilon\eta)} \langle\psi|H_0|\psi\rangle \qquad \mathcal{F}(\rho_*) = \min_{[\psi]\in\mu^{-1}(\rho_*)} \langle\psi|H_0|\psi\rangle$$

Proof idea

$$\mathcal{F}(\rho_*+\epsilon\eta) = \min_{[\psi]\in\mu^{-1}(\rho_*+\epsilon\eta)} \langle\psi|H_0|\psi\rangle \qquad \mathcal{F}(\rho_*) = \min_{[\psi]\in\mu^{-1}(\rho_*)} \langle\psi|H_0|\psi\rangle$$

Intuition: the fibers $\mu^{-1}(\rho_* + \epsilon\eta)$ and $\mu^{-1}(\rho_*)$ are "$\sqrt{\epsilon}$-similar"

Proof idea

$$\mathcal{F}(\rho_*+\epsilon\eta) = \min_{[\psi]\in\mu^{-1}(\rho_*+\epsilon\eta)} \langle\psi|H_0|\psi\rangle \qquad \mathcal{F}(\rho_*) = \min_{[\psi]\in\mu^{-1}(\rho_*)} \langle\psi|H_0|\psi\rangle$$

Intuition: the fibers $\mu^{-1}(\rho_* + \epsilon\eta)$ and $\mu^{-1}(\rho_*)$ are "$\sqrt{\epsilon}$-similar"

μ is the composition:

$$\mathbb{P}(\mathcal{H}) \to \Delta^\Omega \xrightarrow{\mu_{\mathsf{cls}}} \mathrm{conv}(\Omega)$$

1. projection onto standard simplex over Ω:

$$\mathbb{P}(\mathcal{H}) \to \Delta^\Omega, [\psi] \mapsto \|\Pi_\omega\psi\|^2$$

2. *classical density map*:

$$\mu_{\mathsf{cls}} : \Delta^\Omega \to \mathrm{conv}(\Omega), t \mapsto \sum_{\omega\in\Omega} t_\omega \omega$$

Proof idea

$$\mathbb{P}(\mathcal{H}) \to \Delta^{\Omega} \xrightarrow{\mu_{\mathsf{cls}}} \mathrm{conv}(\Omega)$$

Proof idea

$$\mathbb{P}(\mathcal{H}) \to \Delta^{\Omega} \xrightarrow{\mu_{\mathsf{cls}}} \mathrm{conv}(\Omega)$$

- We can "rectify" the fibers of μ_{cls} :

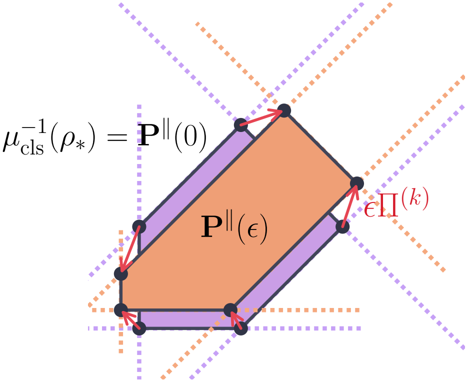
$$\mu_{\mathsf{cls}}^{-1}(\rho_* + \epsilon\eta) \;\approx\; \mathbf{P}^{\|}(\epsilon) \times \mathbf{P}' \;\approx\; \mathbf{P}^{\|}(0) \times \mathbf{P}'$$

Proof idea

$$\mathbb{P}(\mathcal{H}) \to \Delta^{\Omega} \xrightarrow{\mu_{\mathsf{cls}}} \mathrm{conv}(\Omega)$$

▶ We can "rectify" the fibers of μ_{cls} :

$$\mu_{\mathsf{cls}}^{-1}(\rho_* + \epsilon\eta) \;\approx\; \mathbf{P}^{\|}(\epsilon) \times \mathbf{P}' \;\approx\; \mathbf{P}^{\|}(0) \times \mathbf{P}'$$



(only works if ρ_* is regular)

Proof idea

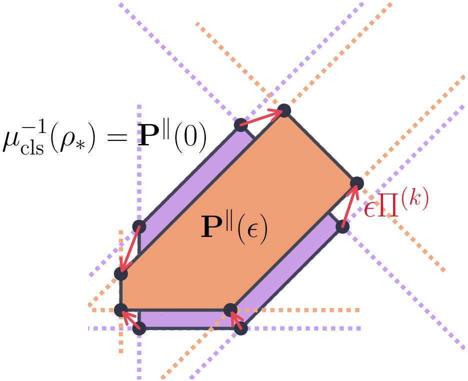
$$\mathbb{P}(\mathcal{H}) \to \Delta^{\Omega} \xrightarrow{\mu_{\mathsf{cls}}} \mathrm{conv}(\Omega)$$

▶ We can "rectify" the fibers of μ_{cls} :

$$\mu_{\mathsf{cls}}^{-1}(\rho_* + \epsilon\eta) \;\approx\; \mathbf{P}^{\parallel}(\epsilon) \times \mathbf{P}' \;\approx\; \mathbf{P}^{\parallel}(0) \times \mathbf{P}'$$

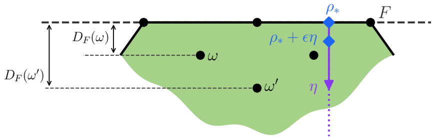
$$\Rightarrow \mu^{-1}(\rho_* + \epsilon\eta) \;\approx\; (S^1)^{\Omega} \times \mathbf{P}^{\parallel}(0) \times \mathbf{P}'$$

phases



(only works if ρ_* is regular)

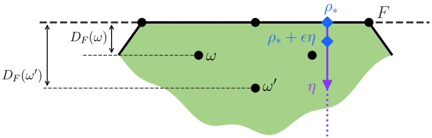
Proof idea



Proof idea

- We can put constrained search into this form:

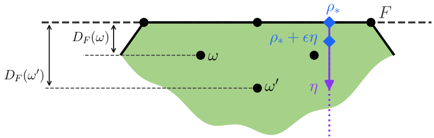
$$\mathcal{F}(\rho_* + \epsilon\eta) = \min_{x \in X} Q(\epsilon, x)$$



Proof idea

- We can put constrained search into this form:

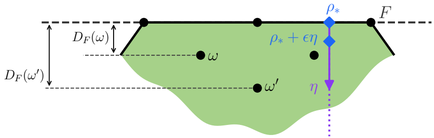
$$\mathcal{F}(\rho_* + \epsilon\eta) = \min_{x \in X} Q(\epsilon, x)$$



- Apply Danskin's theorem:

$$\frac{d}{d\sqrt{\epsilon}}\bigg|_{\epsilon=0} \mathcal{F}(\rho_* + \epsilon\eta) = \min_{x \in \tilde{X}} \frac{d}{d\sqrt{\epsilon}}\bigg|_{\epsilon=0} Q(\epsilon, x)$$

Proof idea



- We can put constrained search into this form:

$$\mathcal{F}(\rho_* + \epsilon\eta) = \min_{x \in X} Q(\epsilon, x)$$

- Apply Danskin's theorem:

$$\frac{d}{d\sqrt{\epsilon}}\bigg|_{\epsilon=0} \mathcal{F}(\rho_* + \epsilon\eta) = \min_{x \in \tilde{X}} \frac{d}{d\sqrt{\epsilon}}\bigg|_{\epsilon=0} Q(\epsilon, x)$$

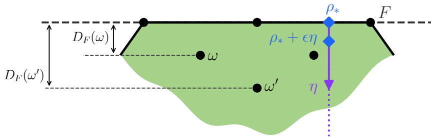
$$Q = Q_{\text{facet}-\text{facet}} + Q_{\text{facet}-\text{off-facet}} + Q_{\text{off-facet}-\text{off-facet}}$$

only this term survives the derivative!

Proof idea

- We can put constrained search into this form:

$$\mathcal{F}(\rho_* + \epsilon\eta) = \min_{x \in X} Q(\epsilon, x)$$



- Apply Danskin's theorem:

$$\frac{d}{d\sqrt{\epsilon}}\bigg|_{\epsilon=0} \mathcal{F}(\rho_* + \epsilon\eta) = \min_{x \in \tilde{X}} \frac{d}{d\sqrt{\epsilon}}\bigg|_{\epsilon=0} Q(\epsilon, x)$$

$$Q = Q_{\text{facet}-\text{facet}} + Q_{\text{facet}-\text{off-facet}} + Q_{\text{off-facet}-\text{off-facet}}$$

only this term survives the derivative!

$\tilde{X} = \cdots \times \mathbf{P}'$, minimization over $\mathbf{P}'$ is easy

$\ldots\ldots \square$

Application - recovering the interaction

Application - recovering the interaction

Definition

For $\mathcal{F}, \mathcal{F}' : \mathrm{conv}(\Omega) \to \mathbb{R}$, define $\mathcal{F} \sim \mathcal{F}'$ if $\exists$ open neighborhood S of the vertices of $\mathrm{conv}(\Omega)$ such that $\mathcal{F}|_S \equiv \mathcal{F}'|_S$.

Application - recovering the interaction

Definition

For $\mathcal{F}, \mathcal{F}' : \mathrm{conv}(\Omega) \to \mathbb{R}$, define $\mathcal{F} \sim \mathcal{F}'$ if $\exists$ open neighborhood S of the vertices of $\mathrm{conv}(\Omega)$ such that $\mathcal{F}|_S \equiv \mathcal{F}'|_S$.

Take two interactions H_0, H_0', let $\mathcal{F}^{(H_0)}, \mathcal{F}^{(H_0')}$ be the respective functionals.

Application - recovering the interaction

Definition

For $\mathcal{F}, \mathcal{F}' : \mathrm{conv}(\Omega) \to \mathbb{R}$, define $\mathcal{F} \sim \mathcal{F}'$ if $\exists$ open neighborhood S of the vertices of $\mathrm{conv}(\Omega)$ such that $\mathcal{F}|_S \equiv \mathcal{F}'|_S$.

Take two interactions H_0, H_0', let $\mathcal{F}^{(H_0)}, \mathcal{F}^{(H_0')}$ be the respective functionals.

Question: if $\mathcal{F}^{(H_0)} \sim \mathcal{F}^{(H_0')}$, what can we say about H_0 and H_0'?

Application - recovering the interaction

Application - recovering the interaction

Investigate this in spin FT:

Application - recovering the interaction

Investigate this in spin FT:

- $\mathcal{H} = (\mathbb{C}^2)^{\otimes N}$

Application - recovering the interaction

Investigate this in spin FT:

- $\mathcal{H} = (\mathbb{C}^2)^{\otimes N}$
- $V = \mathbb{R}^N$

Application - recovering the interaction

Investigate this in spin FT:

- $\mathcal{H} = (\mathbb{C}^2)^{\otimes N}$
- $V = \mathbb{R}^N$
- $\iota : V \to \mathrm{Herm}(\mathcal{H}), v \mapsto \sum_{i=1}^{N} v_i Z_i$

Application - recovering the interaction

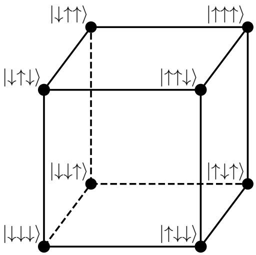
Investigate this in spin FT:

- $\mathcal{H} = (\mathbb{C}^2)^{\otimes N}$
- $V = \mathbb{R}^N$
- $\iota : V \to \mathrm{Herm}(\mathcal{H}), v \mapsto \sum_{i=1}^N v_i Z_i$
- H_0 e.g. $\sum_{i\sim j} X_i X_j$

Application - recovering the interaction

Investigate this in spin FT:

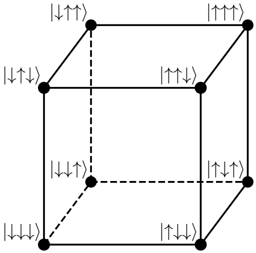
- $\mathcal{H} = (\mathbb{C}^2)^{\otimes N}$
- $V = \mathbb{R}^N$
- $\iota : V \to \mathrm{Herm}(\mathcal{H}), v \mapsto \sum_{i=1}^{N} v_i Z_i$
- H_0 e.g. $\sum_{i\sim j} X_i X_j$



Application - recovering the interaction

Investigate this in spin FT:

- $\mathcal{H} = (\mathbb{C}^2)^{\otimes N}$
- $V = \mathbb{R}^N$
- $\iota : V \to \mathrm{Herm}(\mathcal{H}), v \mapsto \sum_{i=1}^N v_i Z_i$
- H_0 e.g. $\sum_{i\sim j} X_i X_j$

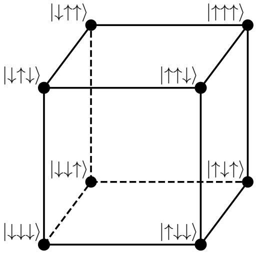


Weights: $\Omega = \{-1, 1\}^N \Rightarrow \mu(\mathbb{P}(\mathcal{H})) = \mathrm{conv}(\Omega) = [-1, 1]^N$

Application - recovering the interaction

Investigate this in spin FT:

- $\mathcal{H} = (\mathbb{C}^2)^{\otimes N}$
- $V = \mathbb{R}^N$
- $\iota : V \to \mathrm{Herm}(\mathcal{H}), v \mapsto \sum_{i=1}^N v_i Z_i$
- H_0 e.g. $\sum_{i\sim j} X_i X_j$



Theorem

If $\mathcal{F}^{(H_0)} \sim \mathcal{F}^{(H_0')}$, then

- $\forall \alpha \in \{\pm 1\}^N : \langle \phi_\alpha, H_0 \phi_\alpha \rangle = \langle \phi_\alpha, H_0' \phi_\alpha \rangle$
- $\forall \alpha, \beta \in \{\pm 1\}^N : |\langle \phi_\alpha, H_0 \phi_\beta \rangle| = |\langle \phi_\alpha, H_0' \phi_\beta \rangle|$

ϕ_α : spin configuration state

$e_{\alpha_1} \otimes \cdots \otimes e_{\alpha_N}$

Application - recovering the interaction

We cannot expect to recover all of H_0:

Application - recovering the interaction

We cannot expect to recover all of H_0:

Definition

For $H_0, H_0' \in \mathrm{Herm}(\mathcal{H})$, define $H_0 \sim H_0'$ if there exists unitary $U \in \mathrm{U}(\mathcal{H})$ such that:

- $\forall v \in V : [U, \iota(v)] = 0$
- $H_0' = U H_0 U^{-1}$

Application - recovering the interaction

We cannot expect to recover all of H_0:

Definition

For $H_0, H_0' \in \mathrm{Herm}(\mathcal{H})$, define $H_0 \sim H_0'$ if there exists unitary $U \in \mathrm{U}(\mathcal{H})$ such that:

- $\forall v \in V : [U, \iota(v)] = 0$
- $H_0' = U H_0 U^{-1}$

$$H_0 \sim H_0' \Rightarrow \mathcal{F}^{(H_0)} \equiv \mathcal{F}^{(H_0')}$$

Application - recovering the interaction

We cannot expect to recover all of H_0:

Definition

For $H_0, H_0' \in \mathrm{Herm}(\mathcal{H})$, define $H_0 \sim H_0'$ if there exists unitary $U \in \mathrm{U}(\mathcal{H})$ such that:

- $\forall v \in V : [U, \iota(v)] = 0$
- $H_0' = U H_0 U^{-1}$

$H_0 \sim H_0' \Rightarrow \mathcal{F}^{(H_0)} \equiv \mathcal{F}^{(H_0')}$

Set $\mathcal{R}(H_0) := \{H_0' \in \mathrm{Herm}(\mathcal{H}) \mid \mathcal{F}^{(H_0)} \sim \mathcal{F}^{(H_0')}\} / \sim$

Application - recovering the interaction

We cannot expect to recover all of H_0:

Definition

For $H_0, H_0' \in \mathrm{Herm}(\mathcal{H})$, define $H_0 \sim H_0'$ if there exists unitary $U \in \mathrm{U}(\mathcal{H})$ such that:

- $\forall v \in V : [U, \iota(v)] = 0$
- $H_0' = U H_0 U^{-1}$

$$H_0 \sim H_0' \Rightarrow \mathcal{F}^{(H_0)} \equiv \mathcal{F}^{(H_0')}$$

Set $\mathcal{R}(H_0) := \{H_0' \in \mathrm{Herm}(\mathcal{H}) \mid \mathcal{F}^{(H_0)} \sim \mathcal{F}^{(H_0')}\}/\sim$

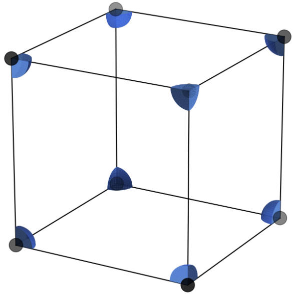
Theorem

For spin functional theories, $|\mathcal{R}(H_0)| < \infty$ for almost all interactions $H_0 \in \mathrm{Herm}(\mathcal{H})$.

Application - recovering the interaction

Theorem

For spin functional theories, $|\mathcal{R}(H_0)| < \infty$ *for almost all interactions* $H_0 \in \mathrm{Herm}(\mathcal{H})$.



Summary & Outlook

Summary & Outlook

- Abelian functional theories

Summary & Outlook

- Abelian functional theories
- Boundary force formula for abelian functional theories

Summary & Outlook

- Abelian functional theories
- Boundary force formula for abelian functional theories
- Consequence: interaction recoverability in spin systems

Summary & Outlook

- Abelian functional theories
- Boundary force formula for abelian functional theories
  - How about general (nonabelian) functional theories?
- Consequence: interaction recoverability in spin systems

Schilling, C., Benavides-Riveros, C. L., Vrana, P. Reconstructing quantum states from single-party information. Phys. Rev. A. (2017)

Summary & Outlook

- Abelian functional theories
- Boundary force formula for abelian functional theories
  - How about general (nonabelian) functional theories?
- Consequence: interaction recoverability in spin systems
  - Most likely can be generalized ...

Schilling, C., Benavides-Riveros, C. L., Vrana, P. Reconstructing quantum states from single-party information. Phys. Rev. A. (2017)

Summary & Outlook

- Abelian functional theories
- Boundary force formula for abelian functional theories
  - How about general (nonabelian) functional theories?
- Consequence: interaction recoverability in spin systems
  - Most likely can be generalized ...
- Functional approximation scheme?

Schilling, C., Benavides-Riveros, C. L., Vrana, P. Reconstructing quantum states from single-party information. Phys. Rev. A. (2017)

Schilling group LMU LUDWIG-MAXIMILIANS-UNIVERSITÄT MÜNCHEN $\mapsto$ UNIVERSITÉ DE GENÈVE

Postdoc/PhD positions

e.g., *BeyondDFT* erc Consolidator Grant

Thank you!

Wang, C.-C., Schilling, C. A Geometric Approach to Strongly Correlated Bosons: From *N*-Representability to the Generalized BEC Force. Phys. Rev. B. (forthcoming)

in preparation...

Penz, M., Liebert, J., **Wang, C.-C.**, Schilling C. What is a Functional Theory?

Wang, C.-C., Schilling, C. On a Universal Boundary Behavior of Generalized Density Functional Theories.

Regular and critical densities

A density $\rho \in F$ is *regular* if ρ is not in the convex hull of $\dim F$ weights.